# lecture 13. COMPOSITE.

$(a,b) \in R$ $\qquad A \times B$

$(b,c) \in S$ $\qquad B \times C$.

$(a,c) \in S \circ R$ if $(a,b) \in R \wedge (b,c) \in S$.

Ex 20 / 405 :-

$R = \{(1,1),(1,4),(2,3),(3,1),(3,4)\}$.

$S = \{(1,0),(2,0),(3,1),(3,2),(4,1)\}$

$S \circ R = \{(1,0),(1,1),(2,1),(2,2),(3,0),(3,1)\}$.

$R \circ S \neq S \circ R$.

$R \circ R = R^2$

$R^2 \circ R = R^3$

$R^3 \circ R = R^4$

$\vdots$

$R^{n-1} \circ R = R^n$

## N-ARY RELATIONS.

$A = \{a,b\}$.

$A \times A \times A = \{(a,a,a),(a,a,b),(a,b,a),(a,b,b),(b,a,a),(b,a,b),(b,b,a),(b,b,b)\}$.

Ex 1. $R = \{(a,b,c) \mid a < b < c\}$

$N \times N \times N$.

$(2,5,3) \in R$. ✗

$(1,2,3) \in R$. ✓

**Ex 2.** $R = \{(a,b,c) \mid b = a+k \wedge c = a+2k\}$.

$\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}$

$k \in \mathbb{Z}$.

$(1,3,5) \in R$. ✓.

$3 = 1 + k$ $\qquad$ $5 = 1 + 2.2$.

$\Rightarrow k = 2$. $\qquad$ $5 = 5$ ✓.

$(2,5,9) \in R$. ✗.

$5 = 2 + k$ $\qquad$ $9 = 2 + 2.3$.

$\Rightarrow k = 3$. $\qquad$ $9 = 2+6$

$9 \neq 8$.

**Ex 3** $R = \{(a,b,m) \mid a \equiv b \bmod m\}$

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$(14,0,7) \in R$ ? ✓.

$(-1,9,5) \in R$. ✓

Ex 4. $A \times N \times S \times D \times T$.

A = Set of Airlines.

N = " " Flight Numbers.

S = " " Departure City.

D = " " Destination ".

T = " " Departure Times.

Table. $\rightarrow$ (PIA, PK712, PEW, KHI, 1500).

| | A | N | S | D | T. |
|---|---|---|---|---|---|
| → | PIA | PK712 | PEW | KHI | 1500. |

Relational Database.

Matrices. [ ]

Size = Rows x Col.

$M = [m_{ij}]$

rows Col.

## Representing Relations Using Matrices.

$R$ $\quad A = \{a_1, a_2, \ldots a_m\}.$

$B = \{b_1, b_2, \ldots b_n\}.$

$M_R.$ rows $= |A| = m$

Col $= |B| = n.$

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R. \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

**Ex1** :- $A = \{\overset{a_1}{1}, \overset{a_2}{2}, \overset{a_3}{3}\}. \quad B = \{\overset{b_1}{1}, \overset{b_2}{2}\}.$

Ex1 :-
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$A = \{1, 2, 3\}$. $B = \{1, 2\}$.

$R = \{(2,1), (3,1), (3,2)\}$.

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

$$\begin{array}{c|cc} & 1 & 2 \\ \hline 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{array}$$

$R = \{(a,b) \mid \overset{a > b.}{a+b = 3}\}$. $A = \{1, 2, 3, 4\}$.

$$M_R = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 1 & 1 & 0 & 0 \\ 4 & 1 & 1 & 1 & 0 \end{array}.$$

Quiz 4. 10-OCT-2025.

$R = \{(a,b) \mid a \subseteq b \lor a \cap b \neq \phi\}$. $A = \{\{a\}, \{a,b\}, \{c,d\}, \{d\}\}$.

Find $M_R$.
Relation in Matrix form.

Solution.

$\{a\}$ $\{a,b\}$ $\{c,d\}$, $\{d\}$

$$\begin{array}{l c} \begin{array}{l} \{a\}. \\ \{a, b\}. \\ \{c, d\}. \\ \{d\}. \end{array} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$